Formation and Growth of Amorphous Colloidal $CaCO_3$ Precursor Particles as Detected by Time-Resolved SAXS

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Time-resolved synchrotron small-angle X-ray scattering (SAXS) studies were performed to investigate the unseeded formation and growth of colloidal calcium carbonate particles. Equimolar aqueous solutions of $CaCl_2 \cdot 2H_2O$ and Na_2CO_3 were rapidly mixed in a stopped-flow apparatus, and SAXS data were recorded using an image-intensified CCD detector. It is shown that SAXS allows studying those processes in situ, with a very good time resolution. It can provide unsurpassed real-time information about the particle size, shape, polydispersity, inner structure, and density. In these studies, well-defined, spherical $CaCO_3$ particles with colloidal dimensions up to ca. 270 nm and a remarkable uniformity in size could be observed. After a short nucleation period, the number density of the growing spheres remains constant. From the evaluation of the absolute scattering intensities, the particle mass density could be determined to be ca. 1.62 g/cm^3, which is considerably lower than the density of the crystalline modifications. Our data thus point to the formation of colloidal, amorphous particles that are a precursor modification of the thermodynamically stable calcite. It was found that these particles are isolated and do not form larger aggregates. Upon lowering the concentration of the educts, particle formation and growth are considerably slowed and smaller particles are being formed.

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Introduction

The formation of solid, colloidal calcium carbonate ($CaCO_3$) particles from a supersaturated aqueous salt solution is a process of considerable geochemical, biological, and industrial importance. It has attracted research for more than a century, recently with a renewed interest in the field of biomineralization.[1−3] Numerous efforts have been made[4−13] in order to understand which polymorph, particle size, shape, surface charge, and so forth forms under certain experimental conditions. It was found that a large number of parameters such as the degree of supersaturation, temperature, mixing conditions, impurities, additives, solvent, and so forth need to be considered. The formation of $CaCO_3$ involves the simultaneous and rapid occurrence of nucleation and growth as well as secondary processes such as aging and agglomeration.[9] This requires the consideration of complex chemical reaction equilibria and of additional diffusion processes.[8,14−16] These processes are difficult to separate and investigate independently. Furthermore, one has to differentiate the cases of primary nucleation, which may be homogeneous or catalyzed by foreign particles, and secondary nucleation, which is initiated by the presence of preformed crystals of the crystallizing phase itself.[9] In case of primary crystallization, it was found that at low supersaturations, heterogeneous nucleation will be dominant, and at high supersaturations, homogeneous nucleation will occur.[11] Due to the complexity of those processes, a comprehensive understanding of the precipitation of $CaCO_3$ has not been achieved so far.[13]

The thermodynamic driving force for a reactive crystallization from solution is given by the difference in chemical potential of the crystallizing compound in a supersaturated solution and in the crystal. The degree of supersaturation[9] is the key variable in any precipitation and governs the rate of nucleation and growth. There exist three crystalline, anhydrous polymorphs of $CaCO_3$, namely, calcite, aragonite, and vaterite. In addition, a crystalline monohydrate and a hexahydrate are known. The observation of a transient amorphous modification has also been reported,[8,10,14,17−19] but this material has not been thoroughly characterized yet. Under ambient conditions, calcite is the only thermodynamically stable modification; all others are metastable. According to Ostwald's rule of stages,[9] however, the metastable phases of higher solubility may be formed first during a precipitation process.

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The system then initially remains supersaturated with respect to the more insoluble phases, which are also being formed. Once the solution concentration decreases below the saturation value with respect to the precursor, this modification undergoes a solid-phase transformation or redissolves and transforms stepwise into a more stable phase, and then into a final reaction product. This stepwise formation of various polymorphs during the course of precipitation of calcium carbonate has been monitored by several groups.[4,7,8,10−12,18,20]

A variety of experimental methods have been applied to gain information about the formation, growth, and precipitation of $CaCO_3$.[20] For real-time investigations on the kinetics, turbidimetry,[11] dynamic light scattering (DLS),[4,21] electrochemical ion activity measurements,[8,10,22,23] and pH monitoring[10,22,24] were used. Scanning and transmission electron microscopies (SEM, TEM)[4,7,10,11,20,25] and optical[8] and X-ray[18] microscopies allow studying the particle shape and morphology. Crystalline modifications were determined by X-ray powder diffraction,[5,10,19] electron diffraction,[10] IR spectroscopy,[8] and polarizing optical microscopy.[8] The widely used SEM and TEM techniques allow producing visible images of the particles, but they are prone to artifacts. In addition, those measurements are time-consuming and leave some ambiguity[19] about whether the particles that are seen on the images really originally existed in solution or rather precipitated later from the supersaturated, drying solution. Light scattering and electrochemical methods have the advantage that they can be applied in situ with a very good time resolution, when used in combination with a rapid mixing chamber. On the other hand, they cannot provide much real-time information about the particle size, shape, uniformity, or morphology. The particle dimension obtained by DLS is a hydrodynamic quantity that may differ greatly from the actual size because the particles carry charges on their surfaces. X-ray microscopy can be applied in situ, but the time resolution is poor. It is thus obvious that a single experimental in situ method that allows study of the fast kinetics even at high supersaturation and which at the same time yields real-time information about particle size and morphology would greatly facilitate more comprehensive investigations and lead to a better understanding.

Small-angle X-ray scattering (SAXS)[26−28] is a well-established technique to characterize the size, shape, inner structure, and interaction of colloids. Thanks to the availability of high-brilliance X-ray beams at synchrotron radiation facilities, kinetic SAXS measurements have become feasible with a time resolution in the millisecond regime.[29,30] In contrast to electron or X-ray microscopies, SAXS data represent an average over a very large number of colloidal particles because all particles located within the illuminated volume contribute to the scattering. This method thus seems to be ideally suited to investigate the formation and growth of colloidal $CaCO_3$ particles. Surprisingly, however, no such reports have appeared in the scientific literature up to now. A few static SAXS measurements on colloidal $CaCO_3$ particles dispersed in nonaqueous media have been published,[31−35] but those works are not related to questions that are to be discussed in this paper. On the other hand, time-resolved synchrotron SAXS measurements have already proven to be a powerful tool to study the formation and growth of silica particles.[36−38] Simultaneous, time-resolved SAXS/wide-angle X-ray scattering (WAXS) investigations on the crystal growth in zeolite synthesis were also reported.[39]

In this work, we present a real-time synchrotron SAXS investigation of the formation of calcium carbonate particles starting from an unseeded, highly supersaturated salt solution. The goal is to characterize the initially formed particles with respect to their geometric and inner structures.

Experimental Section

Calcium chloride dihydrate ($CaCl_2 \cdot 2H_2O$) and sodium carbonate (Na_2CO_3) of analytical grade were purchased from Merck and used without further purification. Aqueous solutions (7−9 mM) of both components were prepared using Millipore water, and the pH was adjusted to 10.5 by NaOH.

Synchrotron small-angle X-ray scattering experiments were conducted at the high-brilliance undulator beamline ID2[30] of the European Synchrotron Radiation Facility (ESRF, Grenoble, France). The wavelength λ of radiation was set to 1 Å, and the sample-to-detector distance to 5 and 10 m. The two-dimensional SAXS images were recorded by an image-intensified CCD camera detector.[30] Incident and transmitted beam intensities were monitored using calibrated PIN diodes. The acquisition time for one frame was 50−100 ms. To improve the statistics, all measurements were repeated at least four times and subsequently averaged. This procedure was justified because the reproducibility of the data was very good (see below).

The rapid mixing of the solutions was accomplished by a BioLogic stopped-flow device. It consisted of three motorized syringes (two for the reactants and one for a cleaning agent) and two mixers. The last mixer is coupled to the scattering cell made of a thin-walled quartz capillary with a diameter of 1.5 mm and a wall thickness of ca. 10 μm. The combined dead time for mixing and transferring from the last mixer to the capillary cell is ca. 5 ms for the flow rate (2−4 mL/s) used in these studies. The apparatus was thermostated at 25 °C. Data acquisition was hardware-triggered at the end of the movement of the motor-driven syringe.

Data Analysis

The standard treatment of the experimental data before their analysis included careful detector corrections for flat field response, spatial distortion, and dark current of the CCD, as well as normalization by the incident flux, sample transmission, and exposure time and by the angular acceptance of the detector pixel elements.[30] The scattering

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from the cell filled with pure water was measured separately and subtracted from the scattering pattern of the sample. The resulting isotropic scattering patterns were azimuthally averaged to obtain $I(q)$, which is the intensity scattered by the sample as a function of the scattering vector $q = 4\pi/\lambda \sin(\theta/2)$, where θ is the scattering angle. Absolute scattering intensities, that is, the differential scattering cross section per unit sample volume and per unit solid angle, were obtained by calibration with a lupolen standard.

The SAXS amplitude $B(q)$ of a single particle with spherical symmetry is given by[26]

$$B(q) = 4\pi \int_0^R [\rho(r) - \rho_m] \frac{\sin(qr)}{qr} r^2 \, dr \qquad (1)$$

where R denotes the particle radius, $\rho(r)$ the local electron density within the particle, and ρ_m the electron density of the dispersion medium. For an ensemble of noninteracting spherical particles with a given size distribution, the scattering intensity $I(q)$ is calculated from $B(q)$ as follows:

$$I(q) = \sum_i N_i B_i^2(q) \qquad (2)$$

where the index i refers to the fraction of particles of radius R_i and with a particle number density N_i. In the special case of homogeneous spheres, $I(q)$ then results to

$$I(q) = \sum_i N_i (\rho - \rho_m)^2 V_i^2 \left(3 \frac{\sin(qR_i) - qR_i \cos(qR_i)}{(qR_i)^3} \right)^2 \qquad (3)$$

with V representing the volume per particle. The scattering curve exhibits several oscillations of a characteristic frequency from which the average particle size may be determined. From the depth and number of observable minima of these oscillations, information about the size polydispersity is obtained.[40] As will be discussed further below, a Gaussian particle size distribution was assumed. The scattering intensity is proportional to the square of the excess electron density of the dispersed particles with respect to the surrounding medium (cf. eq 3). The electron density of a substance is directly proportional to its mass density; as the mass density of $CaCO_3$ is relatively high as compared to for example synthetic polymer latex particles,[40] it exhibits an excellent contrast for X-rays. For example, the excess electron density of calcite (mass density, 2.75 g/cm^3)[41] with respect to water amounts to 495 electrons per nm^3 as compared to only 6 electrons per nm^3 for polystyrene (mass density, 1.05 g/cm^3).[40] So the scattering cross section of calcite exceeds that of polystyrene by a factor of ca. 6800. The formation of $CaCO_3$ particles can thus be monitored by SAXS even at very low concentrations and short exposure times, with good counting statistics and with a good signal-to-background ratio.

In eq 3, only intraparticle interferences are taken into account. This is justified because the concentration of the sample under consideration is very low and interferences arising from interparticle correlations may thus be safely neglected. The structure factor $S(q)$ essentially equals unity in the whole angular range under consideration. Before fitting eq 2 to the experimental data, the theoretical curve was convoluted with a Gaussian function (standard deviation, 0.0059 nm^{-1}) to take into account smearing effects due to the finite beam size and due to the spatial resolution of the detector. Further details concerning this point will be given elsewhere.

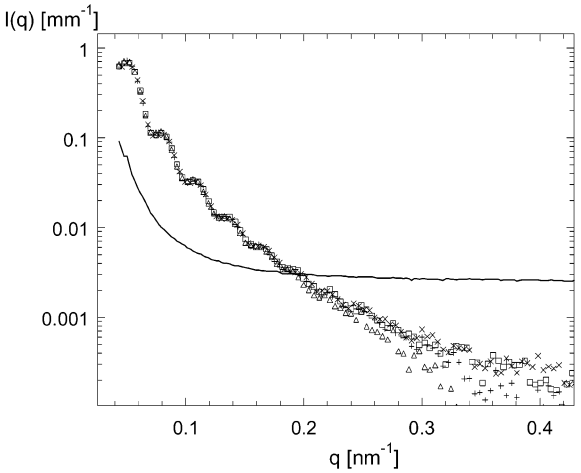


Figure 1. Comparison of the scattering intensities measured from the capillary filled with water (line) and from the system which was prepared by mixing aqueous solutions of 9 mM $CaCl_2 \cdot 2H_2O$ and 9 mM Na_2CO_3 (marks). Those data were collected 9.5 s after mixing with an exposure time of 100 ms. The various marks represent repeated measurements where the scattering from the water-filled capillary was subtracted. For the sake of clarity, only every other data point is shown.

Results and Discussion

To test the reliability and significance of the experimental SAXS data, stopped-flow measurements were repeated several times and checked for their reproducibility. As a representative example, Figure 1 compares four scattering curves which were collected in repeated stopped-flow measurements 9.5 s after equal volumes of equimolar (9 mM) aqueous $CaCl_2 \cdot 2H_2O$ and Na_2CO_3 solutions were mixed. The exposure time for each measurement was 100 ms only. Note that the background scattering originating from the water-filled capillary was subtracted from those displayed data. Evidently, the highly characteristic scattering curve could be very well reproduced in the angular range where the scattering intensity from the sample itself exceeded or equalled that of the background. It is this angular range in which characteristic oscillations in the scattering patterns are observed and from which the most significant structural information may be deduced. The oscillation frequency and amplitude, the overall decay, and the height of the scattering curves could thus be safely monitored with a very good time resolution. By averaging over repeated measurements, the reliability and statistics of the experimental data could be further improved, in particular also at the higher scattering angles. The very good reproducibility suggests that dust particles play no significant role in particle formation. This may be understood by the fact that dust particles are outnumbered by several orders of magnitude by the nanoparticles that are being formed (see below). Furthermore, the SAXS intensity from macroscopic dust particles is confined to the region of the very smallest angles, which in practice is not accessible. Dust particles thus do not interfere with our SAXS measurements, which is a great advantage over any light scattering method.

The time evolution of the SAXS intensities measured from an equimolar mixture of $CaCl_2 \cdot 2H_2O$ and Na_2CO_3 solutions, with the salt concentration in each solution

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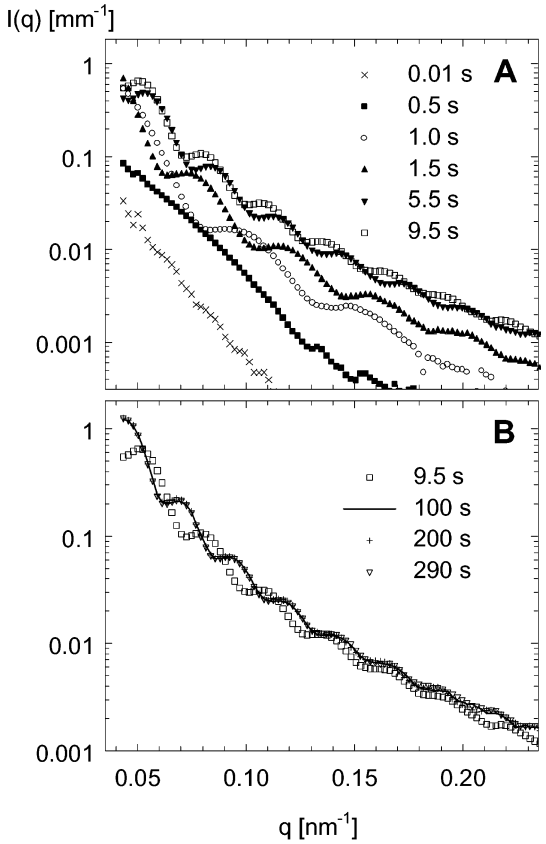


Figure 2. Time-resolved SAXS data measured after rapidly mixing aqueous solutions of 9 mM $CaCl_2 \cdot 2H_2O$ and 9 mM Na_2CO_3 in a stopped-flow apparatus. The legend displays the point of time t at which data collection was started. The exposure time was 50 ms for $t \leq 1.5$ s, and 100 ms for $t > 1.5$ s. After a short induction period of less than 500 ms, the scattering patterns exhibit marked oscillations. Particle growth is evidenced from the increase in the oscillation frequency and the concomitant increase of the intensity (A). After 100 s, no further changes in the scattering patterns were observed (B). For the sake of clarity, only a few selected curves are shown.

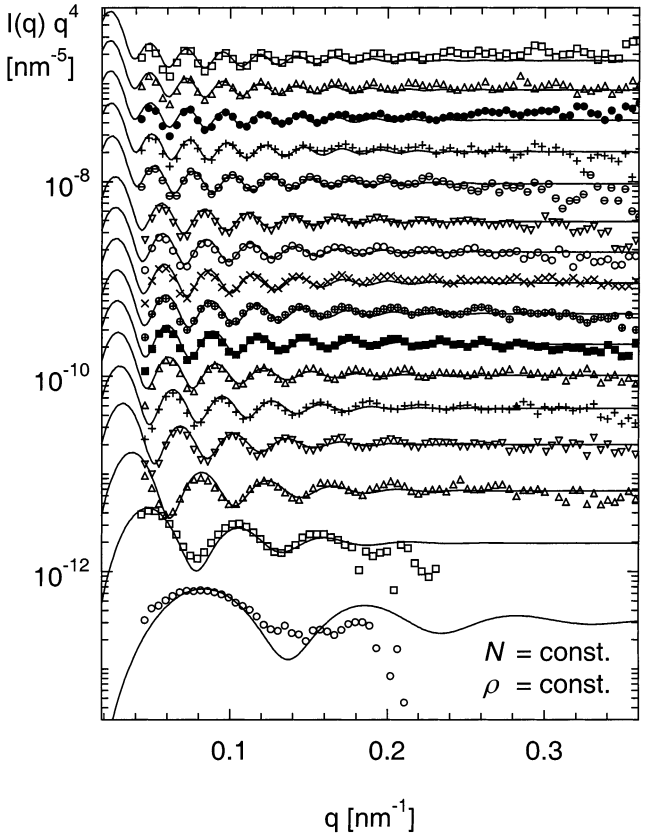


Figure 3. $I(q)\,q^4$ versus q plot of the experimental, time-resolved SAXS data (marks) measured after rapidly mixing equal volumes of aqueous, equimolar solutions (9 mM) of $CaCl_2 \cdot 2H_2O$ and Na_2CO_3. For the sake of clarity, the intensities of consecutive curves were offset by a factor of 2 relative to each other ($t = 0.5$ s is the reference where no offset was used) and only every other data point is shown. The solid lines are the respective fit curves which were calculated for a system consisting of growing polydisperse spheres (number average radius R, polydispersity σ) with a homogeneous electron density ρ within their entire volume. All experimental data were fitted with the same particle number density N and with the same ρ. t represents the point of time at which the measurement was started. The respective fit curves (from bottom to top) were obtained with the following parameters [t, R, σ]: 0.5 s, 32 nm, 11%; 1.0 s, 57.0 nm, 11%; 1.5 s, 75.0 nm, 8%; 2.5 s, 91.5 nm, 7%; 3.5 s, 99.0 nm, 6%; 4.5 s, 103.5 nm, 6%; 5.5 s, 106.0 nm, 5%; 6.5 s, 107.5 nm, 5%; 7.5 s, 109.5 nm, 5%; 8.5 s, 111.5 nm, 5%; 9.5 s, 113.0 nm, 5%; 20.0 s, 125.0 nm, 4%; 40.0 s, 129.0 nm, 4%; 70.0 s, 132.0 nm, 4%; 200.0 s, 133.0 nm, 4%; 280.0 s, 133.0 nm, 4%.

amounting to 9 mM/L, is displayed in Figure 2. For sake of clarity, only a few representative curves are shown. A rapid increase of the SAXS intensities could be observed especially within the first 20 s after mixing (cf. Figure 2A), which shows that particles with colloidal dimensions are being formed. The scattering curves exhibit several oscillations, and the frequency of those oscillations increases with time. Simultaneously, the number of observable fringes increases. This reflects the growth of the particles with a concomitant decrease of their size polydispersity. After 100 s, no further changes could be observed (cf. Figure 2B), and data collection was terminated after 300 s. Obviously particle growth ended after 100 s and a (probably metastable) equilibrium condition was reached. As the overall scattering intensity also remained constant, the settlement of the formed particles can be excluded. We have also measured SAXS data with an increased sample-to-detector distance of 10 m which allows exploration of smaller scattering vectors down to $q = 0.022$ nm^{-1}. These curves at the smallest angles allow detection of larger aggregates of nanoparticles, which would manifest themselves in a significant upturn of the curves. Our data exhibit no such characteristics, which shows that the sample consists of essentially isolated nanoparticles.

When displaying the data from Figure 2 in an $I(q)\,q^4$ versus q plot (cf. Figure 3), one observes that all curves level off at the higher scattering angles and run parallel to the q-axis. This shows that the intensities decay in inverse proportion to the fourth power of the scattering vector. According to Porod,[42] this is a characteristic scattering feature of well-defined, three-dimensional particles with sharp interfaces. Also shown in Figure 3 are fit curves that were calculated using eq 3 together with the fitted values for the number-average particle radius R and for the size polydispersity σ. The good agreement between the experimental data and the fit curves clearly shows that the system may be well described as an ensemble of growing, homogeneous spherical particles.

Already during the first 500 ms, particles with an average radius of ca. 32 nm have been formed. After 1.5 s, the spheres have grown to a radius of 75 nm and exhibit a remarkably narrow size distribution with a standard

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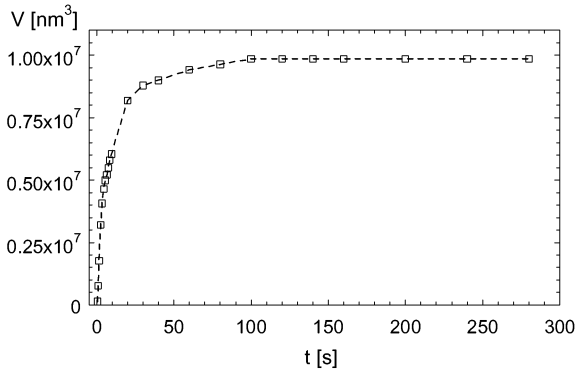


Figure 4. Increase of the volume V per particle as a function of time (initial salt concentrations, 9 mM).

deviation of only 8%. At the point of time when particle growth ended ($t = 100$ s), the particle radius has reached 133 nm and the polydispersity further decreased to ca. 4%. The rapid initial increase of the particle volume followed by a period of slower growth until reaching the equilibrium is visualized in Figure 4. For modeling the experimental data, we have assumed a Gaussian particle size distribution. On the other hand, it was suggested by others[43] that growing calcium carbonate particles exhibit a log-normal distribution. We performed model calculations using either distribution in order to estimate how the scattering curves are affected. It was found that for a standard deviation below 10% the differences between both distributions are only marginal so that they may not be differentiated in our experiments.

From eq 3, it can be seen that SAXS data also contain information about the number density N and about the electron density ρ of the colloidal particles. As pointed out above, the absolute scattering intensity depends on both of these interdependent parameters. It is particularly noteworthy that the complete set of experimental data (from $t = 0.5$ s to $t = 290$ s) could be fitted with the same number density N and with the same electron density ρ of the particles. It is thus suggested that nucleation was virtually completed within the first 500 ms and that in the following time the particles are growing without a significant change of their density. Our observation of a very narrow particle size distribution also corroborates the finding that nucleation occurs very rapidly within a short induction period.

For the determination of N and ρ, we proceeded as follows: using the fitted maximum particle size that was reached at equilibrium ($R = 133$ nm), one may calculate N by taking into account the known initial concentrations and by assuming that the final particle size corresponds to 100% conversion. In a first approximation, the water solubility of $CaCO_3$ is neglected. N also depends on the mass density of the formed $CaCO_3$ particles. The density may be chosen from tabulated values[41,44] for calcite (2.75 g/cm^3), aragonite (2.95 g/cm^3), vaterite (2.65 g/cm^3), or the hexahydrate (1.77 g/cm^3). For the monohydrate and for the amorphous modification, no data were found in the literature. If one would assume that an anhydrous, crystalline modification has formed, the calculated scattering intensity exceeds the measured one by a factor of 11−14. Assuming the formation of the hexahydrate, which has the lowest known density of all crystalline polymorphs, there is still a discrepancy by a factor of 3. These significant discrepancies clearly indicate that no such nanocrystalline

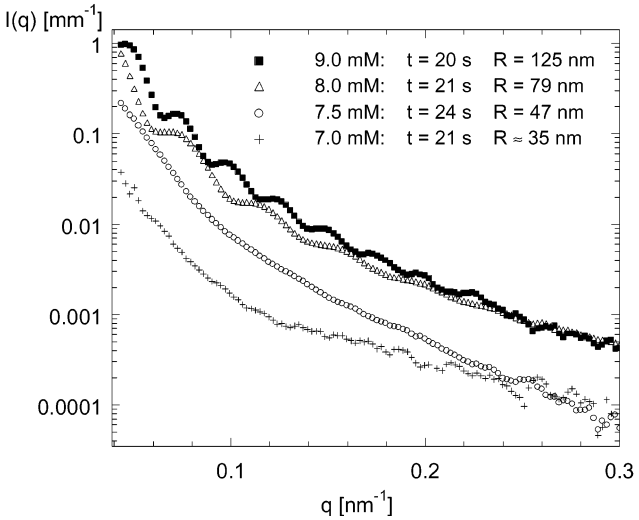


Figure 5. Comparison of the scattering curves that were obtained after rapidly mixing equimolar (7−9 mM) solutions of $CaCl_2 \cdot 2H_2O$ and Na_2CO_3. The respective measurements were started at about the same point of time ($t = 22 \pm 2$ s) at which supersaturation was achieved. The legend gives the initial concentrations, the point of time t, and the fitted particle radius R.

modifications had formed in our experiments and that the actual density is considerably lower than that of the crystalline modifications.

When the mass density of the particles is taken as a free fit parameter (with the water solubility of $CaCO_3$ still being neglected), a value of 1.49 g/cm^3 is obtained. It is thus very likely that this corresponds to an amorphous, low-density modification of calcium carbonate, which could also include a certain amount of water.[14] Within the framework of Ostwald's rule of stages (see above), this may be well understood as a metastable precursor modification of the thermodynamically stable calcite. Our data show that those amorphous nanospheres are isolated, well-defined particles with a very narrow size distribution.

The formation of amorphous calcium carbonate (ACC) requires that the ion activity product of the initially supersaturated solution is higher than the thermodynamic solubility product k_s of ACC. Independent investigations by Brecevic et al.[14] and by Clarkson et al.[8] came to the same conclusion that ACC has a well-defined value for k_s: from their precipitation−dissolution experiments using turbidimetry, they determined the pk_s ($= -\log(k_s)$) value of ACC at 25 °C to be 6.393 and 6.04, respectively. These values are considerably lower than those reported for the crystalline polymorphs calcite (8.480), aragonite (8.336), and vaterite (7.913)[14] but comparable to that of $CaCO_3 \cdot 6H_2O$ (6.62).[8] From the pk_s value, Brecevic et al. calculated the solubility of ACC in pure water to be 1.7 mM. Thus, the initial supersaturation with respect to ACC in our experiment can be estimated to be 2−3 and the formation of an ACC precursor, as was suggested from our experimental data, is corroborated. The water solubility of ACC is non-negligible compared to the initial concentration of 4.5 mM for Ca^{2+} and CO_3^{2-} ions in our own experiments and should thus be taken into account in our analysis.

From a revised fitting procedure, in which the water solubility of ACC was taken into account, the mass density of ACC is then determined from our SAXS data to be 1.62 g/cm^3 and the particle number density N to be 1.75×10^{10} cm^{-3}. Upon lowering the initial salt concentrations, we observed a marked retardation of particle growth and the formation of smaller particles. This retardation is dem-

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(44) *Handbook of Chemistry and Physics*, 47th ed.; Weast, R. C., Selby, S. M., Eds.; CRC Press: Cleveland, OH, 1966−1967.

onstrated in Figure 5, which compares the scattering curves where data collection started at about the same point of time and where the initial concentrations were varied: about 22 s after mixing, the particles had grown to an average radius of 125, 79, 47, and ca. 35 nm for initial concentrations of 9, 8, 7.5, and 7 mM, respectively.

The results of our work are consistent with what has been reported by others and can additionally provide new insights into the formation and properties of ACC. The observation of ACC that forms from a supersaturated solution was reported by some other groups; its physicochemical properties could not be thoroughly determined, though. One reason is its short lifetime, as it readily transforms into more stable modifications within a short period of time. Ogino et al.[10] reported the observation of unstable ACC that forms spherical particles with a diameter of ca. 100 nm. This result was based on measurements using an ion sensitive electrode in combination with X-ray diffraction and electron microscopy. Clarkson et al.[8] and Kabasci et al.[7] also stated the formation of an ACC precursor, but no detailed information about particle size and shape could be given. Furthermore, Rieger et al.[18] discussed the formation of a transient, amorphous modification. ACC is also abundant in biominerals, and there is increasing evidence that organisms synthesize organic macromolecules specifically for stabilizing ACC with respect to the thermodynamically stable phase calcite.[1,2,17] The stabilization of ACC for extended periods of time by specially designed dendrimers was reported by Donners et al.[45]

(45) Donners, J. J. J. M.; Heywood, B. R.; Meijer, E. W.; Nolte, R. J. M.; Roman, C.; Schenning, A. P. H. J.; Sommerdijk, N. A. J. M. *Chem. Commun.* **2000**, (19), 1937.

Conclusion

This work has demonstrated for the first time the potential of time-resolved synchrotron SAXS to study in situ the formation and growth of $CaCO_3$ from supersaturated salt solutions. By use of a single experimental method, comprehensive information about the size, shape, modification, mass density, number density, and aggregation behavior of the formed $CaCO_3$ particles could be obtained as a function of the reaction time. Evidence for the formation of amorphous calcium carbonate, which is a metastable precursor of calcite, could be given. It was shown that the number of the growing spherical particles and their mass density remain constant after a short nucleation period has passed. We plan further SAXS experiments to study the mechanism of particle formation during the earliest stages and to investigate the transitions between the various polymorphs.

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